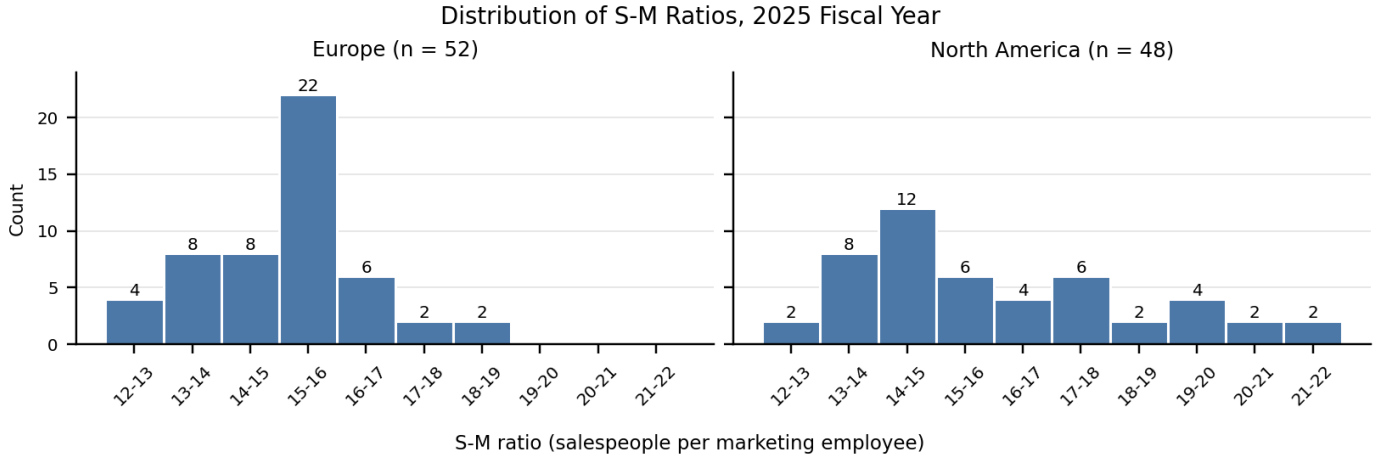
SOCS Comparison Practice: S-M Ratios in Europe and North America

Show all your work. Indicate clearly the methods you use, because you will be graded on the correctness of your methods as well as on the accuracy of your results and explanation.

A multinational corporation calculates the **sales-to-marketing personnel ratio (S-M ratio)**, defined as the **number of salespeople per marketing employee**, for each of its branches. The histograms show the S-M ratios for the **52 branches in Europe** and the **48 branches in North America** during the 2025 fiscal year.



Region	n	Mean	SD	Min	Q1	Median	Q3	Max
Europe	52	15.12	1.37	12.3	14.2	15.3	15.9	18.3
North America	48	16.06	2.36	12.6	14.4	15.3	17.1	21.8

Summary statistics are calculated from the raw data. **Extreme values:** Europe — two smallest 12.3, 12.5; two largest 18.1, 18.3. North America — two smallest 12.6, 12.9; three largest 20.6, 21.3, 21.8.

Task (20 points): Compare the distributions of S-M ratios for the branches in Europe and North America. Address **shape, outliers, center, and spread**, justify your answers with appropriate calculations, and **interpret your results in context**.